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RESEARCH

Molecular Assessments of Antimicrobial Protein Enterocins and Quorum Sensing Genes and Their Role in Virulence of the Genus *Enterococcus*

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Abstract
Enterococcus has emerged as an opportunistic pathogen because of its antibiotic resistance and virulence profile, which makes it a causative agent of several diseases like endocarditis, surgical site, and urinary tract infections. Currently, species of this genus are the 2nd most frequently isolated microorganisms from hospital-acquired infections. Significant association with hospitals and unhygienic conditions of the environments has made them resistant to a wide range of antibiotics. On the brighter side, enterococci have the ability to produce antimicrobial proteins (i.e., enterocins) that exhibit wide antagonistic activity, thus making them useful microbes in the food and pharmaceutical industries. Enterocins are also involved in niche control in gut microbiota which is regulated by the quorum sensing (QS) system. A bacterial communication system that is controlled by the *fsr* operon in enterococci consists of *FsrABDC*, *ef1097*, and *GelE/SprE* genes. Hence, the present study was conducted for molecular assessment of enterocins and quorum sensing genes, inter-environmental correlation, and species prevalence of enterococci isolated from different environmental niches of Karachi, Pakistan. Obtained results revealed the highest prevalence of *E. faecium* and *E. faecalis* in all environments. Bacterial antagonism and enterocin genes were observed significantly high in poultry environments. The inter-environmental correlation indicated a strong positive correlation of freshwater with sewage and soil environments. Similarly, the *fsr* regulatory system was mostly identified in poultry-related environments, and a significant correlation between QS system and biofilm formation was established. In conclusion, this study confirmed the high prevalence of *E. faecium* in all tested sources, high enterocin production in non-clinical environments, and more *fsr* regulatory genes in poultry-related environments.

Keywords *Enterococcus* · Enterocins · Enterolysin-A · Quorum sensing · Sex pheromones · Virulence

Introduction

The genus *Enterococcus* becomes an increasingly pathogenic microorganism which was previously considered a common commensal of the gut [1–3]. This genus has a ubiquitous distribution and can withstand harsh environmental conditions [4]. Their widespread distribution is mostly supported by their genomic plasticity and low GC contents, due to which 58 species are currently identified [5]. Characteristically, these microorganisms are homo fermentative, oxidase and catalase negative [6], aero tolerant, and growing in high salinity and pH [7], and show resistance to desiccation, UV radiation, heavy metals, and starvation [3, 8, 9]. These microbes also help in the organoleptic properties of foods and are used in many biotechnological, pharmaceutical, and food industries, and several strains are also used as probiotics [10, 11]. Due to the distinguishing behavior of enterococci, they are also used as environmental fecal indicator bacteria (FIB) [8, 12, 13]. Being an opportunistic pathogen, this genus shows association with hospital-acquired infections (HAIs) [14] and acts as a causative agent of many diseases like wound, intra-abdominal and urinary tract infections (UTI), meningitis, and endocarditis [4, 15, 16]. The two major species responsible for 70–80% of these complications are *E. faecalis* and *E. faecium*, and mostly identified from hospital sources [7, 8, 17].

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The presence of a dozen different types of virulence traits increases their pathogenic nature and limits their useful applications [7–9, 15, 18]. These traits are causing severe complications, for instance, hemolysin can disrupt the RBCs, gelatinase can cleave the gelatin and other related proteins, sex pheromones are involved in the formation of aggregation substances, biofilm former strains enhanced the enterococcal pathogenesis, aggregation substances help in bacterial colonization and facilitate the transfer of virulence and antibiotic resistance genes, and DNAse can degrade the host DNA [1, 4, 15, 19–23]. Cumulatively, these pathogenic determinants make the genus *Enterococcus* doubtful and still not get the status of Generally Recognized As Safe (GRAS) and Qualified Presumption of Safety (QPS) [1, 7, 18].

Currently, the antimicrobial resistance profile of enterococci is notorious as they show resistance to the last-resort antibiotics like vancomycin and linezolid, and hence become a global challenge for healthcare providers [4, 20, 24–26]. Unfortunately, both intrinsic and acquired antibiotic resistance are identified in enterococcal strains from different origins with high prevalence in hospitals and poultry-related environments [4, 7]. Among the various enterococcal species, *E. faecalis* considered the most virulent, while *E. faecium* is more associated with antibiotic resistance [9].

On the brighter side, enterococci can produce multiple antimicrobial bacteriocins, termed as enterocins [6, 22]. These are ribosomally synthesized low molecular weight peptides having broad and narrow spectrum antimicrobial activities. Characteristically, these are heat stable, hydrophobic, 20–60 amino acid residue peptides with self-defense system [1, 3, 27]. Different enterocins are identified including enterocin A, enterocin B, enterocin P, enterocin Q, enterocin L50(L50A/B), enterocin ef1097, enterocin SE-K4, and enterocin AS-48 which not only impart the biotechnological properties but also give the anti-cancer potential to the enterococcal species [28–30]. Enterocins are classified into three main classes. Class-I are small and low molecular weight peptides having extensive post-translational modifications [27, 31], while Class-II are unmodified heat-stable peptides having strong antilisterial activity [6]. Class-III enterocins are high molecular weight, heat-labile proteins and are generally known as bacteriolysins, e.g., enterolysin-A [32].

Enterocin production is regulated by a phenomenon called quorum sensing (QS) which is a population density sensing mechanism in bacteria through which intra and inter-bacterial species communication occurs in their close vicinity via the production of auto-inducer (AI) peptides [9, 21, 33]. This communication occurs in both Gram-positive and Gram-negative bacteria and plays a significant role in bacterial life [34]. Mechanistically, auto-inducers (i.e., small molecules having the capability to cross cell membrane) are released during bacterial growth and bind with two-component sensor kinases which initiate bacterial communication by the transcription of related genes [9, 35, 36]. This phenomenon provides the bacterial species to respond according to the cell density and synchronize accordingly via the regulation of gene expression [9, 33].

The quorum sensing system of *E. faecalis* is regulated by *fsr* regulatory system which consists of four genes, i.e., *fsr-ABCD* [37], and involved in the extracellular accumulation of the *fsrD*-encoded gelatinase biosynthesis-activating pheromone (GBAP) [21]. *FsrB* converts the precursor *fsrD* into the mature GBAP, which is then released extracellular. Once the histidine kinase *fsrC* detects this accumulation, it activates *fsrA* by phosphorylating [19, 37]. The transcription of *fsrA* up-regulates the expression of three main target loci (i.e., *fsrBCD*, *gelE-sprE* operons, and ef1097). Thus, the *fsr* regulatory system regulates the expression of the *gelE* and *sprE* proteases, which in turn control virulence, biofilm formation, and protein turnover [2, 21]. Quorum sensing pathways are now considered an alternative target for antimicrobial treatment [38] because cells in biofilms can communicate with each other via QS and most of the clinical species use this system to regulate the process of virulence development [23], as documented that the *fsr* quorum sensing is also involved in the regulation of *gelE* and *sprE* [14, 19, 22].

This study aims to establish the statistical correlation among enterococcal strains isolated from different environmental niches of Karachi, a very densely populated city of Pakistan. Different virulence traits, particularly bacterial antagonism, enterocins (antimicrobial proteins and peptides), *fsr* quorum sensing (regulatory system) genes, and the gene flow among them were elucidated, a research area scarcely addressed in the literature. Obtained results revealed the highest prevalence of *E. faecium* and *E. faecalis* in all environments. Bacterial antagonism and enterocin genes were detected significantly high, particularly in the poultry environments. The inter-environmental correlation indicated a strong positive correlation of freshwater with sewage and soil environments suggesting the alarming hygienic status. Moreover, the *fsr* regulatory system was mostly identified in poultry-related environment, where a significant correlation between QS system and biofilm formation was established.

Materials and Methods

Data and Strain Collections

Previously, our group isolated enterococci from different environmental niches of Karachi, Pakistan from 2014 to 2018 [8, 39–42]. The sources of collection include poultry, sewage, fresh water, agricultural and bulk soil, and clinical environments. All the collected isolates ($n = 2650$) were initially grown on Brain Heart Infusion (BHI) broth (CM 1032, Oxoid) and then on enterococcal-specific media, i.e., Slanetz-Bartley agar (SBA; CM0377A, Oxoid), to identify them as the genus

Enterococcus [40–42]. Various other confirmatory tests like catalase and urease were also performed to confirm their enterococcal origin. For molecular identification, the genus (16S rRNA) and species-specific (*sodA*) gene amplifications [8] based on multiplex PCR strategy were used that also confirm their species level identification [8, 40–42].

Enterococcal Species Prevalence in Different Environments

For the analysis of our previously reported data [8, 39–42], results were critically analyzed and reported the enterococcal species prevalence. The results were retrieved from our previous published data sets and comparatively analyzed for their species prevalence in different environments.

For cross-confirmation, genomic DNA from the selected isolates was extracted by the CTAB method [43] and used as a template for single or multiplex PCR analysis of different virulent traits and characterizations [44]. The primers were reconstituted according to their molar concentrations using Tris–EDTA buffer of pH 8.2 as recently described by Rehman et al. [42]. The genus (16S rRNA) and species (*sodA*) identification of the selected strains were performed by using the optimized PCR reaction conditions as reported earlier [42].

Phenotypic and Genotypic Evaluation of Bacterial Antagonism

The bacteriocinogenic activity was checked against indicator strains *E. faecium* NA283 and *L. monocytogenes* ATCC13932 using stab-overlay and spot-on-lawn methods as described by Hussain et al. [1]. Positive bacterial antagonism potential was considered their extant of zone of inhibitions around the colonies of indicator strains which was measured in mm [1, 39]. The activity was performed in Biosafety Cabinet class II (ESCO, Japan). The molecular assessment of known enterocin genes in enterococcal strains was investigated by using various enterocin gene primers [44] as recently reported by Hussain et al. [1]. Different enterocins like enterocin A, enterocin B, enterocin P, enterocin L50 (*L50A* + *L50B*), enterocin ef1097, enterocin SEK-4, enterocin Q, and enterolysin-A were investigated by single and multiplex PCR, and their products were visualized by using UV trans-illuminator (UVP. Birmingham, UK).

Phenotypic and Genotypic Assessment of Major Virulence Traits

Both secretory and surface virulence factors were investigated both phenotypically by plate assays and genotypically by targeting their associated target genes via single or multiplex PCR by using the protocols previously described by us [1, 39]. These virulence traits include lipase, catalase, gelatinase, and DNase. The genes screened for virulence assessment include sex pheromones (*cob*, *eep*, *ccf*, *cpd*), aggregation substance (*agg*, *asa1*), and adhesion-related (*efaAfm*, *efaAfs*, *ace*, *acm*) genes [44].

Molecular Assessment of Quorum Sensing Genes

Quorum sensing phenomena in bacteria are controlled by *fsr* operon (i.e., *gelE/sprE*, *fsrA*, *fsrB*, and *fsrC*). These genes were also screened by applying single and multiplex PCR. The primer concentration, reaction, and thermal conditions were followed as described earlier by our group [39, 40, 44].

Environmental Correlation of the Enterococcal Strains

The inter-environmental correlation of enterococcal strains identified and rigorously analyzed by our group was statistically analyzed with reference to explore the role of antimicrobial proteins enterocins and quorum sensing genes and their role in virulence of the genus *Enterococcus* by Pearson's correlation plot using Python version 3.9. A heat map of Pearson's correlation plot of all environmental strains was constructed, and significance was reported as $p < 0.05$ value.

Results

Prevalence of Enterococci and Their Species in Different Environments

The enterococcal species isolated from different environments and identified by multiplex PCR strategy (Fig. S1a) varied significantly in their distribution. The maximum number of isolates was collected from the poultry environment ($n = 1358$), followed by soil ($n = 490$) and clinical ($n = 350$) sources. Similarly, sewage and its related environment isolates ($n = 290$) were collected and confirmed as enterococcal species (Fig. 1a).

The phenotypic and molecular assessments of these isolated strains were showing the most widespread prevalence of *E. faecium* and *E. faecalis*. *E. faecium* was highly dominated in poultry environments ($n = 490$) followed by soil ($n = 378$) and sewage ($n = 173$) sources, while *E. faecalis* was highly prevalent in clinical ($n = 303$) and sewage ($n = 93$) sources, followed by poultry ($n = 74$) and soil ($n = 58$) environments. Other enterococcal species like *E. hirae* ($n = 24$), *E. gallinarum*, *E. mundtii*, and *E. casseliflavus* ($n = 11$) were also identified in a lesser extent. *E. mundtii* was more prevailed in poultry isolates ($n = 10$) while *E. hirae* was more detected in soil ($n = 16$) source. Similarly, *E. gallinarum* ($n = 5$) strains were confirmed from both poultry and soil sources, while ($n = 6$) strains was confirmed from poultry environments (Fig. 1b).

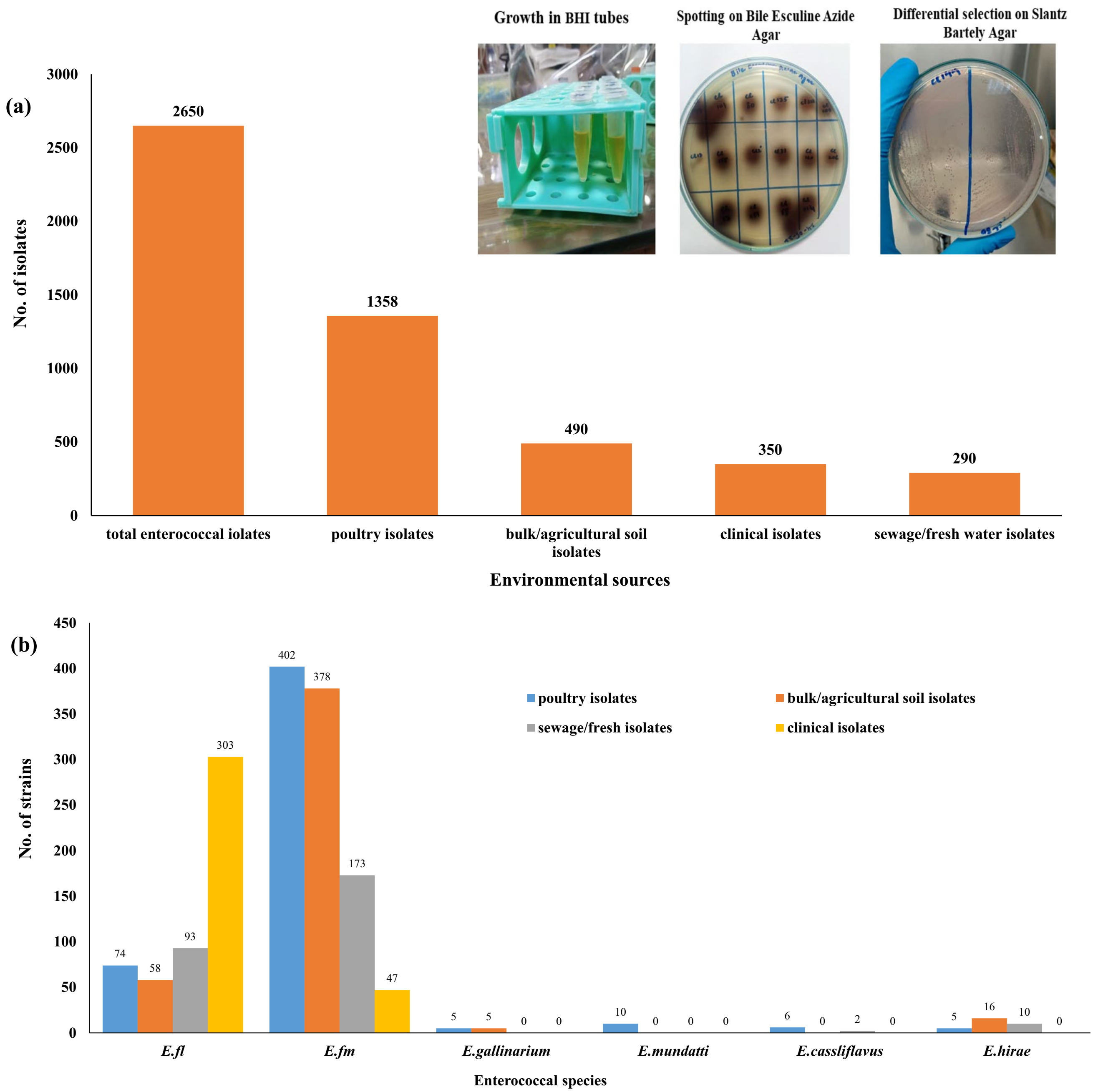

Fig. 1 **a** Isolation of *Enterococcus* species from different environmental niches of Karachi, Pakistan. **b** The specie-wise distribution of enterococcal isolates collected from different environments of Karachi, Pakistan

Bacterial Antagonism in Enterococcal Species

The phenotypic evaluation of bacterial antagonism in all isolated strains from different environments indicated the highest antagonism in poultry and its related environment (40%), followed by soil strains (27.3%) and clinical strains (24.4%). In soil, (agriculture soil) shows more bacterial antagonism (36%) than bulk soil (24%). The lowest bacterial antagonism (7%) was identified in sewage and its related environments as illustrated (Fig. 2).

Molecular Genotyping of Enterocin Genes

The genotypic assessment of bacterial antagonism was conducted by screening various enterocin-producing genes (Fig. S1b) which revealed variable frequencies in different environments and in *Enterococcus* species. Around 60% of *E. faecalis* isolates from agricultural soil show a positive genotype for enterolysin-A (*enlA*), while no *E. faecium* strain from the same source was found positive, rather this source shows positive results for enterocins, including *EntP*

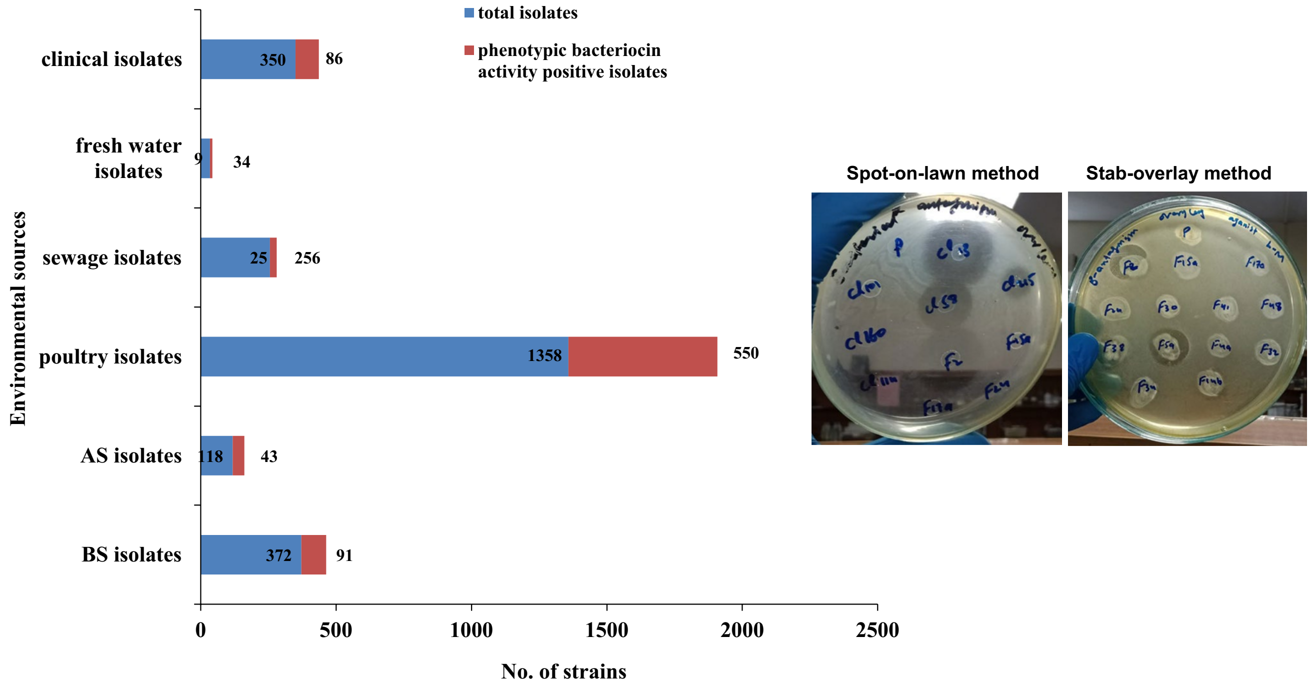

Fig. 2 Phenotypic assessment of bacterial antagonism in enterococcal strains collected from different environments

(29.8%), *EntB* (19.5%), and *EntA* (12.6%). Similarly, in bulk soil, *enlA* was positive only in 18% of *E. faecalis* strains, while only enterocin genes were detected in different percentages in *E. faecium* strains, for instance, *EntP* (10%), *EntA* (8.3%), and *EntB* (6.9%). A very low prevalence of these genes was also observed in *E. faecalis* strains (Fig. 3).

In poultry feces strains, the most detected enterocins were Ent-*ef1097* (90.5%), followed by *EntB* (84.9%) and *EntA* (68%). Likewise, *Ent-SE-K4* (58%), *EntL50* (35.8%), and *EntP* (24.5%) were also detected. Surprisingly, enterolysin-A (*enlA*) was observed in 54.7% of *E. faecium* strains from poultry strains. Multiple enterocin production was also observed in these strains and the combination of *EntA* + *EntB* was detected in 47%, while *EnA* + *EntB* + *EntL50* were also detected in 20% of strains. In the poultry air environment, the most detected enterocins were found in a decreasing order of isolates, *EntB* (85%), *Ent-ef1097* (70%), *EntA* (69%), and *EntSE-K4* (40%). Multiple enterocin production was also observed in poultry air as *EntA* + *EntB* (38%) and *EntA* + *EntB* + *EntL50* in (21.5%) of strains. Surprisingly, the enterolysin-A gene was also detected in *E. faecium* strains with a high prevalence of 45% (Fig. 3). In poultry feed isolates, *Ent-ef1097* and *Ent-SE-K4* were detected in all strains, while *EntB* and *EntA* were detected with 22% and 18%, respectively. The percentage of multiple enterocin production in this environment was observed less as compared to other poultry sources. There was no enterolysin-A gene detected in poultry feed strains. In poultry water source, all screened enterocins were observed in *E. faecalis* strains with the highest prevalence of *EntB* (40%), *Ent-SE-K4* (36%), and *EntA* (32%). Cumulatively, in all poultry sources, almost all enterocins were detected in high percentages in *E. faecium* strains, except *EntA* which is highly detected in *E. faecalis* strains (Fig. 3).

In sewage water, the most detected enterocin was enterolysin-A (17%) followed by *Ent-ef1097* (11%) in *E. faecalis* strains, while *EntP* was highly observed (10%) in *E. faecium* strains. In freshwater, *EntB*, *Ent-ef1097*, and *enlA* were detected with the same frequency (18%) in all *E. faecium* strains, while no enterocin was observed in *E. faecalis* strains. From clinical source, *EntB*, *Ent-ef1097*, and *EntP* were detected with the same percentage (i.e., 20%), while enterolysin-A was only observed in 7% of *E. faecalis* strains (Fig. 3).

Prevalence of Enterococcal Virulence Traits in Different Environments

The prevalence of major virulence traits among all strains collected from different environments was phenotypically analyzed and presented in Fig. 4. The results reflected high hemolytic activity in clinical strains ($n = 190$) followed by poultry environment ($n = 160$). Similarly, it was also observed in agriculture soil ($n = 43$), sewage ($n = 42$), and bulk soil ($n = 18$) strains. The biofilm production was

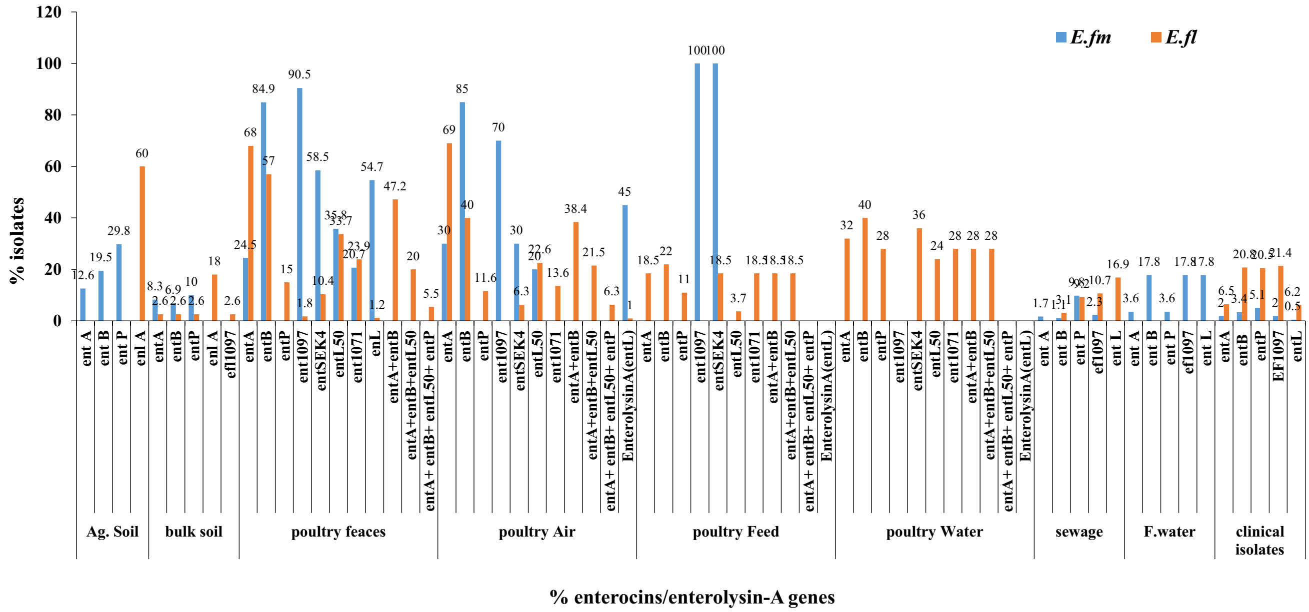

Fig. 3 Comparative genotypic assessment of bacterial antagonism (i.e., enterocin genes) in the collected enterococcal strains from different environments

highly positive in poultry ($n = 324$), followed by clinical ($n = 199$), bulk soil ($n = 182$), and sewage environment ($n = 152$) strains. In comparison, the freshwater strains ($n = 34$) showed moderate biofilm formation potential. DNase activity was highly observed in poultry strains ($n = 36$) followed by sewage ($n = 17$) and clinical ($n = 16$) strains. Likewise, lipase activity was high in the poultry environment ($n = 343$) and then in agriculture soil ($n = 93$) and sewage ($n = 54$) originated strains. Gelatinase production was significantly observed in poultry and its related environment ($n = 110$) followed by clinical ($n = 59$), bulk soil ($n = 31$), sewage ($n = 23$), and freshwater environment

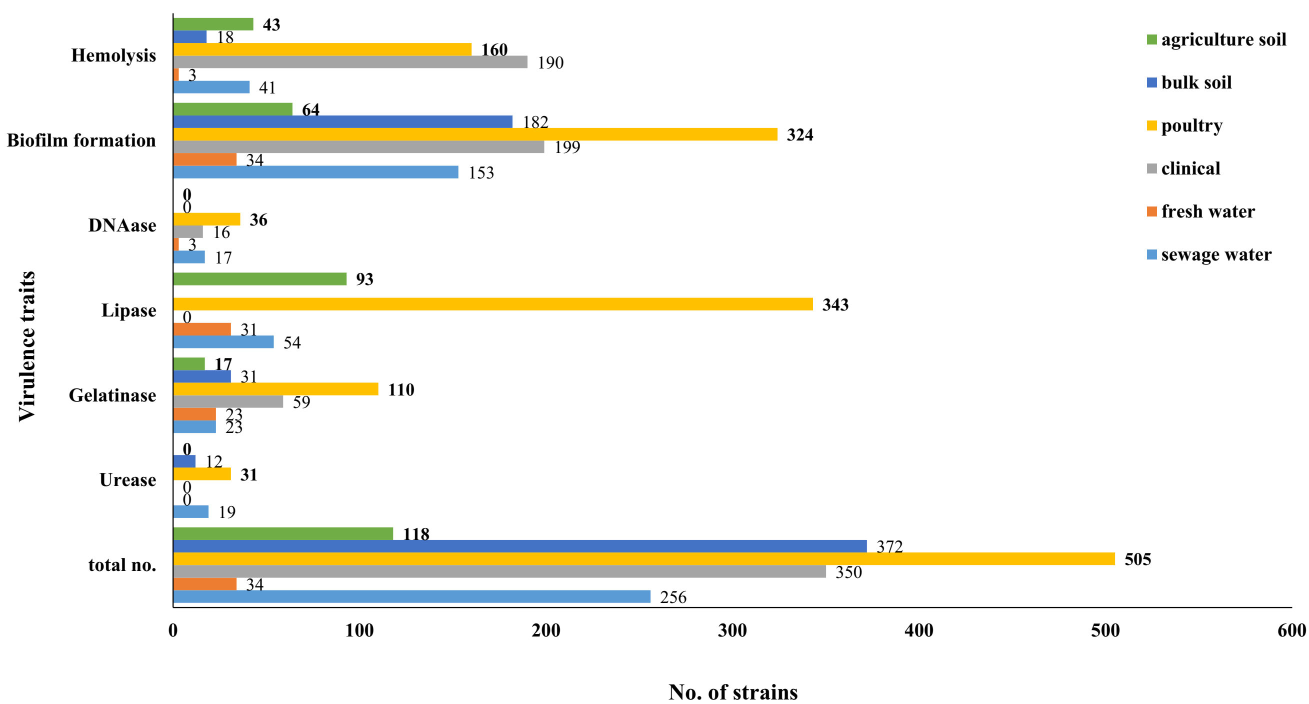

Fig. 4 Comparative phenotypic assessment of different enterococcal virulence traits in enterococcal strains isolated from various environments of Karachi, Pakistan

($n = 23$) strains. The poultry strains also showed a high prevalence of urease production ($n = 31$) followed by sewage ($n = 19$) and bulk soil ($n = 12$) strains (Fig. 4).

Molecular Assessment of Virulence Traits

The molecular assessment of different virulence traits was also screened (Fig. S1c), and their prevalence in different environmental sources was established. *E. faecium* adhesion gene (*EfaAfm*) was highly abundant in bulk soil ($n = 250$) followed by sewage ($n = 92$) strains, while *E. faecalis* adhesion gene (*EfaAfs*) and *espTM* genes were highly detected in clinical strains. Genes that are responsible for aggregation substances, i.e., *asa1*, *ace*, and *agg* were highly detected in clinical strains ($n = 210$, 181, 129, and 26, respectively). Likewise, the *acm* gene was highly prevailed in bulk soil strains followed by sewage source. Sex pheromone responsible genes, i.e., *ccf*, *cpd*, *eep*, and *cob* (Fig. S1d), were also detected but in a lesser extent in all sources, except quite dominated in sewage environment (Fig. 5a).

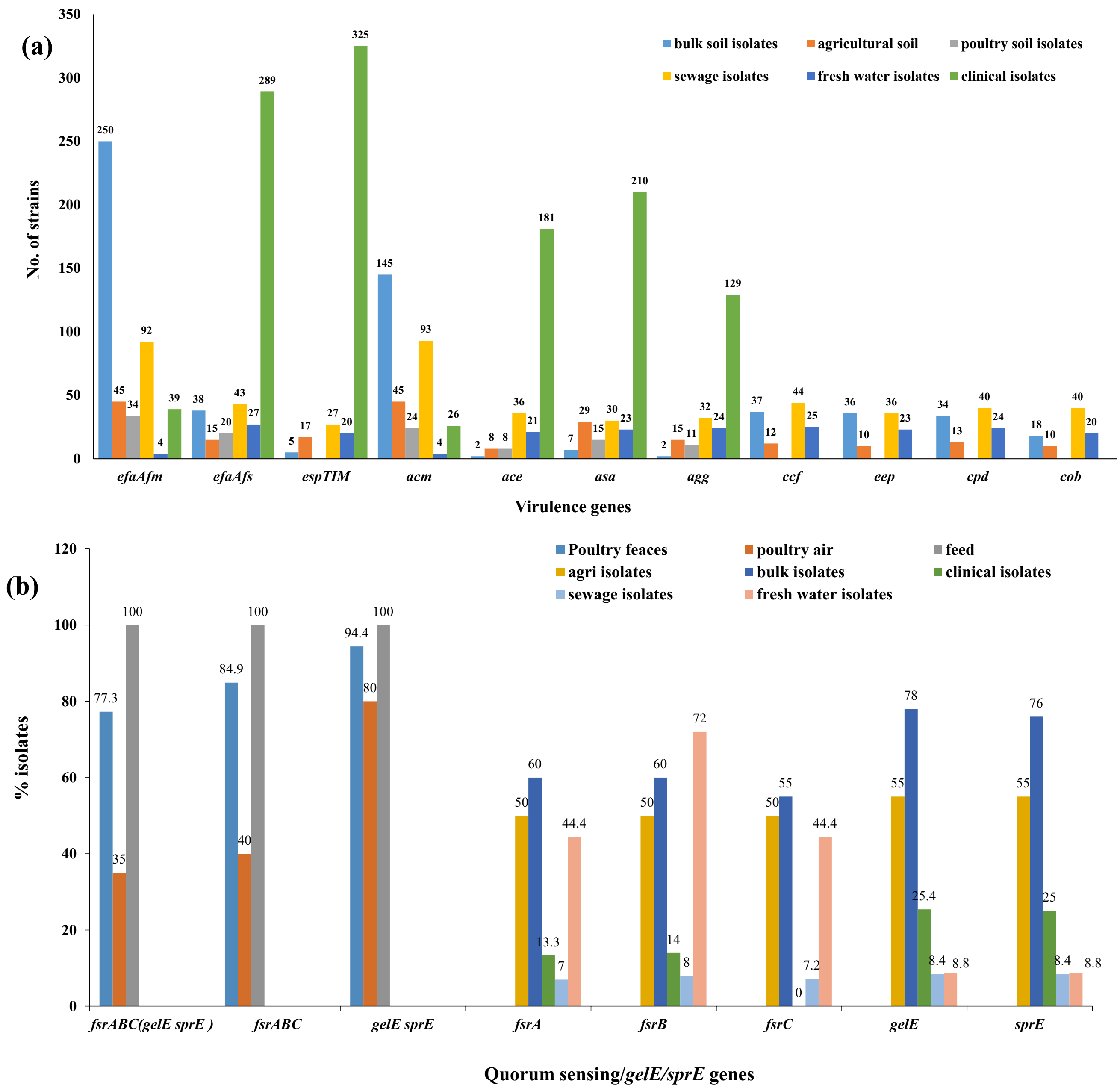

Fig. 5 **a** Molecular assessment of biofilms, aggregation substances, and sex pheromone genes in enterococcal strains isolated from different environments of Karachi, Pakistan. **b** Cumulative graph showing genotyping of quorum sensing genes in enterococcal strains collected from all environments

Molecular Assessment of Quorum Sensing Genes

The genotyping of quorum sensing genes (i.e., *fsr* and *gelE/sprE* operons; Fig. S1c) indicated the highest prevalence in poultry environments including 100% in poultry feeds, 77% in poultry feces, and 35% in poultry air-originated strains. The individual detection and prevalence of these genes were different in all environments, for instance, *fsrA* and *fsrC* were highly detected in bulk soil (60% and 55%, respectively), while *fsrB* was greatly observed in freshwater isolates (72%). Similarly, *gelE* and *sprE* were mostly observed in bulk soil with 78% and 76% (respectively) prevalence. As demonstrated in Fig. 5b, the prevalence of these genes was quite less in clinical and sewage-originated strains.

Environmental Correlation of the Selected Strains

The inter-environmental correlation was statistically analyzed and shown by Pearson's correlation plot. The results indicated the most significant correlation ($p < 0.05$) among poultry feces and freshwater, poultry air, and feed environment. Strains from sewage also show a correlation with poultry feeds and feces and freshwater strains. Likewise, bulk soil and poultry feed correlation was also observed. Surprisingly, there was no correlation observed among clinical strains with other environments as reflected by Pearson's plot (Fig. 6).

Discussion

The genus *Enterococcus* has the ability to tolerate harsh environmental conditions and adaptations due to genome plasticity [1, 12]. From different local environments (i.e., water, sewage, soil, poultry, and clinical) of Karachi, a very densely populated city of Pakistan, a huge data set of 2650 isolates were collected and identified as *Enterococcus* species by our earlier studies [8, 39–42]. The highest number of isolates was collected from poultry and its related

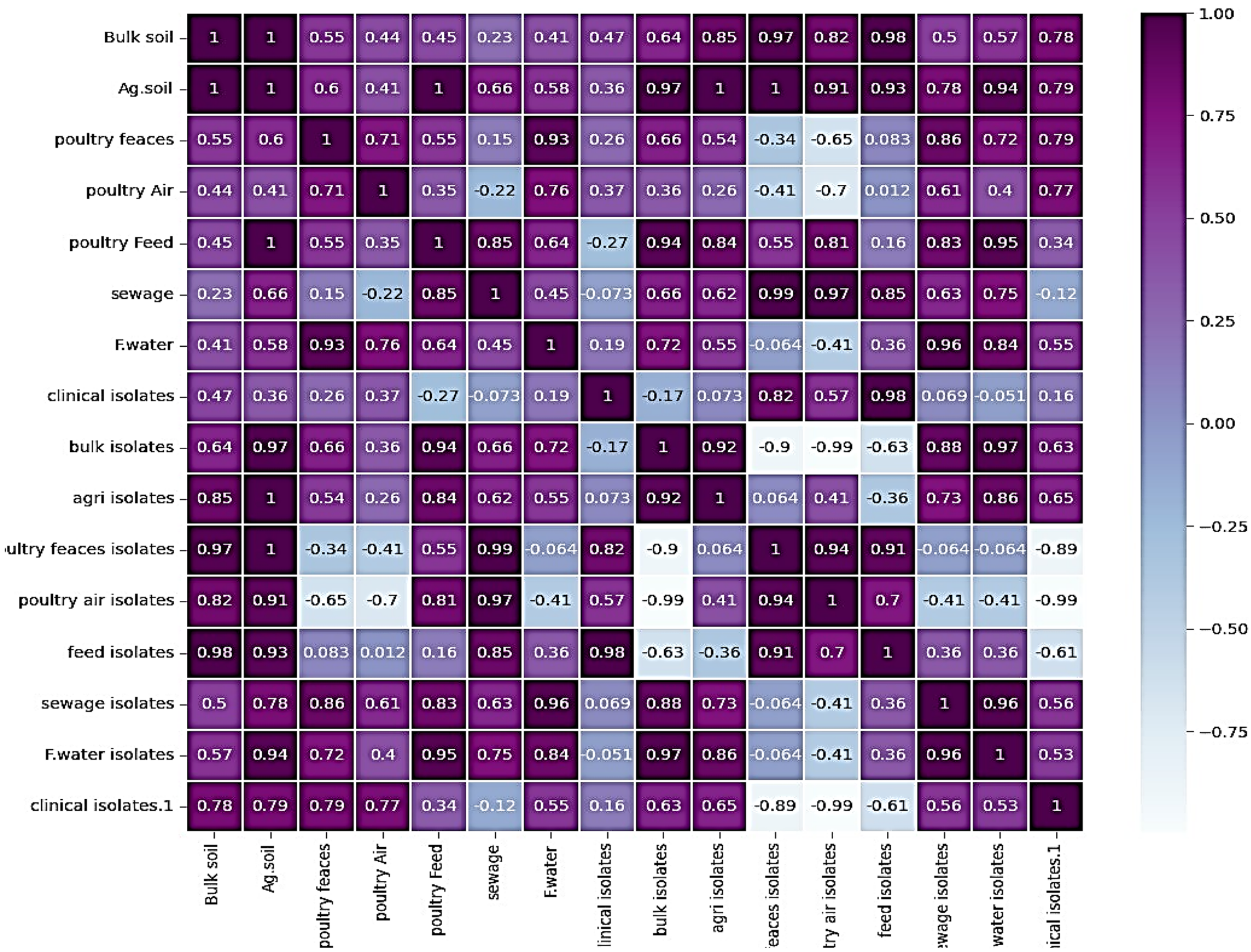

Fig. 6 Pearson's correlation plot of statistically significant values of enterocins/enterolysin-A and *fsrABC* genes in all collected *Enterococcus* species from different environments

environments ($n = 1358$) followed by soil sources ($n = 490$). The high prevalence in these two aforementioned niches may be due to the addition of enterococcal probiotics into the poultry feeds and/or the manure of domestic animals which is used as a natural fertilizer in agricultural fields [45]. At species level, *E. faecium* dominates all other species and highly prevailed in poultry sources, while *E. faecalis* was identified in high numbers from sewage sources. The presence of *IS16* elements in outside clinical environments is also dangerous as this gene is known as a hospital-associated marker [37]. The high prevalence of these two species with varying percentages was also reported by several authors previously [3, 7, 12]. The presence of more bacterial antagonism potential in poultry isolates indicates the extensive usage of *E. faecium* as a feed source in poultry, as generally, *E. faecium* is known for its high bacterial antagonistic activity. Similarly, the high percentage of bacterial antagonism in agriculture isolates also confirms the higher prevalence of *E. faecium* strains.

The enterococcal pathogenic nature can be boosted by the presence of multiple virulence factors that enhance their potential to cause infections. In this study, different virulence traits from various local environments were correlated which indicated their high prevalence in clinical isolates followed by sewage and bulk soil. These results indicated the hospital waste disposal into sewage sources and poor irrigation of the fields, which is quite alarming for the health of humans and animals. It is previously reported that *E. faecalis* isolated from clinical environments have more virulence traits than those isolated from food sources [20], which is also complemented in this study. The phenotypic prevalence of virulence in the aforementioned niches was also confirmed by their molecular genotyping. Studies have also reported the presence of hemolysin and gelatinase production, besides biofilm formation in poultry strains. In addition, biofilm capacity was also found relatively high in clinical strains as previously documented reports show that approximately 25% of enterococcal clinical strains have the potential to cause UTI, wound infection, and endocarditis.

Besides the prevalence of virulence traits in different environments, enterocin production is also considered the key element in enterococcal strains [1]. The comparative analysis of enterocin productions in different environments varies with respect to both sources and species as observed in this study. For instance, *enlA* was the most widespread enterocins in both agricultural and bulk soil observed in *E. faecalis* strains only, while other enterocins like *EntB*, *EntP*, and *EntA* were detected in *E. faecium* strains (Fig. 3). Among poultry sources, single or multiple enterocin genes were detected mostly in *E. faecium* strains. It was amazing to find out the highest prevalence of *entA* in poultry air and feces *E. faecalis* strains, while enterolysin-A was mostly observed in *E. faecium* strains. The most anomalous behavior was shown by poultry water and poultry feeds in which all enterocins were observed in *E. faecalis* strains; only *Ent-ef1097* and *Ent-SE-K4* were observed in *E. faecium* from poultry feeds. Overall in poultry feces, more single enterocin producers were observed in *E. faecium* strains, while multiple producers were mostly detected in *E. faecalis* strains.

In sewage environments, *enlA* and *Ent-ef1097* were only observed in *E. faecalis* strain, while in freshwater, all enterocins including enterolysin-A were detected in *E. faecium* strains. The presence of *enlA* in *E. faecium* strains indicates the contamination of our local environment. Likewise, in clinical environments, *EntA*, *EntP*, and *Ent-ef1097* were found with the same ratios in *E. faecalis*, while in very low percentages in *E. faecium.* In contrast, *enlA* was only identified in *E. faecalis* strains. The role of *enlA* is important because of its proteolytic activity [20, 32].

The quorum sensing and *gelE/sprE* genes were highly detected in poultry feed strains followed by poultry feces. In bulk soil and freshwater strains, *gelE* and *sprE* were observed in high percentages. Due to their protease activities, the presence of such genes in the water environment can make the host unhealthy and more prone to diseases. Interestingly, all these genes were detected in very low percentages in clinical and in sewage strains. Intestinal bacteria can communicate with each other via quorum sensing facilitated by the production of small peptide auto-inducers as they bind with kinases and start the transcription of genes [33]. Previous studies [9, 38] show the relationship between virulence factors and quorum sensing systems. For instance, the *E. faecalis* QS is controlled by the *fsr* regulatory system which in turn regulates *gelE/sprE* gene expression [14, 35, 37]. These proteases are involved in the control of many virulence factors like biofilm formation and gelatinase production [2, 21]. The relationship between biofilm and QS system is important as they help to identify and develop methods for the elimination of pathogenic bacteria, and thus aid to the food safety [33]. Quorum sensing system is used by bacterial communities for communication and also helps in the controlling of virulence trait development especially in *E. faecalis* [23]. In this study, biofilm-associated and gelatinase genes were mostly identified in poultry environments. In a previous study by Ali et al. [8], it was also reported that such virulence genes mostly observed in poultry feces and poultry air environment. Based on the above-presented data and available literature support [2, 9, 11, 14, 19, 23, 35, 37], a positive correlation has also been observed between QS system and other virulence factors, particularly the production of enterocins, proteases, and/or biofilm formation in this study. The summary of this positive intercorrelation between various virulent determinants is also illustrated in Fig. 7.

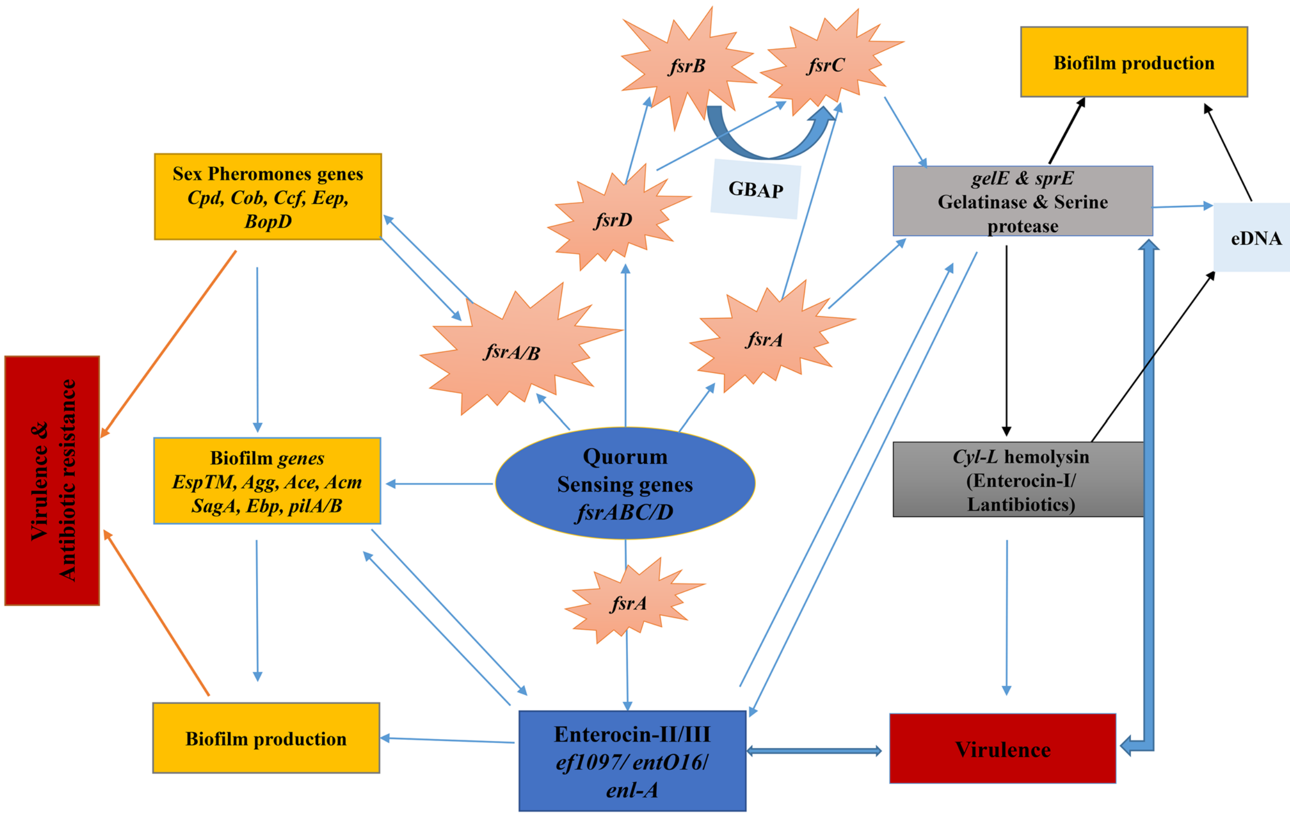

Fig. 7 Schematic representation illustrating the positive inter-correlation between quorum sensing system and other virulence factors, particularly the production of biofilm, proteases, hemolysin, and enterocins

Conclusion

Globally, the pathogenic nature of enterococci is an alarming situation, particularly in the developing world, due to the presence of various virulence determinants, antibiotic resistance, and their dissemination via horizontal gene transfer. However, being ubiquitous in nature and their wide application in many industries, the researchers are working to find a solid correlation among isolates from different environments that help the one-health issues to manage. Herein, a huge data set of strains collected from almost all environmental niches was analyzed, and at the molecular level, their correlation was established that reveal a higher prevalence of enterococci (*E. faecium* and *E. faecalis*) in poultry and sewage environments, while bacterial antagonism and enterocin genes were detected significantly high in the poultry environments. Moreover, the indirect role of antimicrobial proteins and peptides (i.e. enterocins in niche control) and their regulatory machinery (i.e. *fsr* QS system) and their association with other virulent traits present on the same operon (e.g. proteases) is clearly suggested in this study and improvises the need of further in vivo studies.

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Author Contribution Conception, project design, and supervision were performed by SA Ali. Materials preparation, data collection, and analysis were performed by A. Khoso, A. Hussain, S. Akram, D. Ahmed, M. Rehman, H. Bin-Asif, S. Zahid, K. A. Hasan, and S. A. Ali. The manuscript's first draft was prepared by A. Khoso and A. Hussain, and edited and finalized by S. A. Ali.

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Data Availability All data generated or analyzed during this study are included in this published article and the supplementary information available online at the publisher's website.

Declarations

Competing Interests The authors declare no competing interests.

References

1. Hussain A, Akram S, Ahmad D, Rehman M, Ahmed A, Ali SA (2023) Molecular assessment and validation of the selected enterococcal strains as probiotics. Probiotics Antimicrob 1–15. https://doi.org/10.1007/s12602-023-10163-6
2. Pillai SK, Sakoulas G, Eliopoulos GM, Moellering RC Jr, Murray BE, Inouye RT (2004) Effects of glucose on fsr-mediated biofilm formation in *Enterococcus faecalis*. J Infect Dis 190(5):967–970. https://doi.org/10.1086/423139
3. Fisher K, Phillips C (2009) The ecology, epidemiology and virulence of *Enterococcus*. Microbiology 155(6):1749–1757. https://doi.org/10.1099/mic.0.026385-0
4. Fiore E, Van Tyne D, Gilmore MS (2019) Pathogenicity of enterococci Microbiol Spectr 7(4):7–4. https://doi.org/10.1128/9781683670131.ch24
5. García-Solache M, Rice LB (2019) The *Enterococcus*: a model of adaptability to its environment. Clin Microbiol Rev 32(2):10–1128. https://doi.org/10.1128/CMR.00058-18
6. Nigutova K, Morovsky M, Pristas P, Teather RM, Holo H, Javorsky P (2007) Production of enterolysin A by rumen *Enterococcus faecalis* strain and occurrence of enlA homologues among ruminal Gram-positive cocci. J Appl Microbiol 102(2):563–569. https://doi.org/10.1111/j.1365-2672.2006.03068.x
7. Ogier JC, Serror P (2008) Safety assessment of dairy microorganisms: the *Enterococcus* genus. Int J Food Microbiol 126(3):291–301. https://doi.org/10.1016/j.ijfoodmicro.2007.08.017
8. Ali SA, Hasan KA, Bin Asif H, Abbasi A (2014) Environmental enterococci: I. Prevalence of virulence, antibiotic resistance and species distribution in poultry and its related environment in Karachi. Pakistan. Lett Appl Microbiol 58(5):423–432. https://doi.org/10.1111/lam.12208
9. González JE, Keshavan ND (2006) Messing with bacterial quorum sensing. Microbiol Mol Biol 70(4):859–875. https://doi.org/10.1128/mmbr.00002-06
10. Franz CM, Huch M, Abriouel H, Holzapfel W, Gálvez A (2011) Enterococci as probiotics and their implications in food safety. Int J Food Microbiol 151(2):125–140. https://doi.org/10.1016/j.ijfoodmicro.2011.08.014
11. Moreno MF, Sarantinopoulos P, Tsakalidou E, De Vuyst L (2006) The role and application of enterococci in food and health. Int J Food Microbiol 106(1):1–24. https://doi.org/10.1016/j.ijfoodmicro.2005.06.026
12. Byappanahalli MN, Nevers MB, Korajkic A, Staley ZR, Harwood VJ (2012) Enterococci in the environment. Microbiol Mol Biol Rev 76:685–706. https://doi.org/10.1128/mmbr.00023-12
13. Boehm AB (2007) Enterococci concentrations in diverse coastal environments exhibit extreme variability. Environ Sci Technol 41(24):8227–8232. https://doi.org/10.1021/es071807v
14. Pinkston KL, Gao P, Diaz-Garcia D, Sillanpää J, Nallapareddy SR, Murray BE, Harvey BR (2011) The Fsr quorum-sensing system of *Enterococcus faecalis* modulates surface display of the collagen-binding MSCRAMM Ace through regulation of *gelE*. J Bacteriol 193(17):4317–4325. https://doi.org/10.1128/JB.05026-11
15. Yordanova R, Yaneva Z, Gencheva D, Beev G (2022) Antimicrobial resistance distribution and quorum-sensing regulation of enterococcal strains, isolated from hospitalized patients. Appl Sci 12(17):8735. https://doi.org/10.3390/app12178735
16. Liu Y, Wang Y, Schwarz S, Li Y, Shen Z, Zhang Q, Shen J (2013) Transferable multiresistance plasmids carrying *cfr* in *Enterococcus* spp. from swine and farm environment. J Antimicrob Agents 57(1):42–48. https://doi.org/10.1128/AAC.01605-12
17. Akhi MT, Farzaneh F, Oskouei M (2009) Study of enterococcal susceptibility patterns isolated from clinical specimens in Tabriz. Iran Pak J Med Sci 25(2):211–216
18. Bhardwaj A, Kaur G, Gupta H, Vij S, Malik RK (2011) Interspecies diversity, safety and probiotic potential of bacteriocinogenic *Enterococcus faecium* isolated from dairy food and human faeces. World J Microbiol Biotechnol 27:591–602. https://doi.org/10.1007/s11274-010-0494-4
19. Nakayama J, Tanaka E, Kariyama R, Nagata K, Nishiguchi K, Mitsuhata R, Sonomoto K (2007) Siamycin attenuates *fsr* quorum sensing mediated by a gelatinase biosynthesis-activating pheromone in *Enterococcus faecalis*. J Bacteriol 189(4):1358–1365. https://doi.org/10.1128/JB.00969-06
20. Hickey RM, Twomey DP, Ross RP, Hill C (2003) Production of enterolysin A by a raw milk enterococcal isolate exhibiting multiple virulence factors. Microbiology 149(3):655–664. https://doi.org/10.1099/mic.0.25949-0
21. Nakayama J, Yokohata R, Sato M, Suzuki T, Matsufuji T, Nishiguchi K, Sonomoto K (2013) Development of a peptide antagonist against fsr quorum sensing of Enterococcus faecalis. ACS Chem Biol 8(4):804–811. https://doi.org/10.1021/cb300717f
22. Ferchichi M, Sebei K, Boukerb AM, Karray-Bouraoui N, Chevalier S, Feuilloley MG, Zommiti M (2021) *Enterococcus* spp.: is it a bad choice for a good use—a conundrum to solve? Microorganisms 9(11):e2222. https://doi.org/10.3390/microorganisms9112222
23. Li Y, Pan J, Wu D, Tian Y, Zhang J, Fang J (2019) Regulation of *Enterococcus faecalis* biofilm formation and quorum sensing related virulence factors with ultra-low dose reactive species produced by plasma activated water. Plasma Chem Plasma Process 39:35–49. https://doi.org/10.1007/s11090-018-9930-2
24. Ashagrie D, Genet C, Abera B (2021) Vancomycin-resistant enterococci and coagulase-negative staphylococci prevalence among patients attending at Felege Hiwot Comprehensive Specialized Hospital, Bahir Dar. Ethiopia Plos One 16(4):e0249823. https://doi.org/10.1371/journal.pone.0249823
25. Low DE, Keller N, Barth A, Jones RN (2001) Clinical prevalence, antimicrobial susceptibility, and geographic resistance patterns of enterococci: results from the SENTRY Antimicrobial Surveillance Program, 1997–1999. Clin Infect Dis 32(2):S133–S145. https://doi.org/10.1086/320185
26. Schouten, M. A., Hoogkamp-Korstanje, J. A. A., Meis, J. F. G., Voss, A., & European VRE Study Group (2000) Prevalence of vancomycin-resistant enterococci in Europe. Eur J Clin Microbiol 19:816–822. https://doi.org/10.1007/s100960000390
27. Cotter PD, Ross RP, Hill C (2013) Bacteriocins—a viable alternative to antibiotics? Nat Rev Microbiol 11(2):95–105. https://doi.org/10.1038/nrmicro2937
28. Ankaiah D, Palanichamy E, Antonyraj CB, Ayyanna R, Perumal V, Ahamed SIB, Arul V (2018) Cloning, overexpression, purification of bacteriocin enterocin-B and structural analysis, interaction determination of enterocin-A, B against pathogenic bacteria and human cancer cells. Int J Biol Macromol 116:502–512. https://doi.org/10.1016/j.ijbiomac.2018.05.002
29. Franz CM, Stiles ME, Schleifer KH, Holzapfel WH (2003) Enterococci in foods—a conundrum for food safety. Int J Food Microbiol 88(2–3):105–122. https://doi.org/10.1016/S0168-1605(03)00174-0
30. Sharma P, Kaur S, Chadha BS, Kaur R, Kaur M, Kaur S (2021) Anticancer and antimicrobial potential of enterocin 12a from *Enterococcus faecium*. BMC Microbiol 21(1):1–14. https://doi.org/10.1186/s12866-021-02086-5
31. Nilsen T, Nes IF, Holo H (2003) Enterolysin A, a cell wall-degrading bacteriocin from *Enterococcus faecalis* LMG 2333. Appl Environ Microbiol 69(5):2975–2984. https://doi.org/10.1128/AEM.69.5.2975-2984.2003
32. Khan H, Flint SH, Yu PL (2013) Determination of the mode of action of enterolysin A, produced by *Enterococcus faecalis* B9510. J Appl Microbiol 115(2):484–494. https://doi.org/10.1111/jam.12240

33. Fernandes MS, Esper LMR, Kabuki DY, Kuaye AY (2018) Quorum sensing in *Enterococcus faecium*, *Enterococcus faecalis* and *Bacillus cereus* strains isolated from ricotta processing. Ciência Rural Santa Maria 48(2):e20161111. https://doi.org/10.1590/0103-8478cr20161111
34. Whiteley M, Diggle SP, Greenberg EP (2017) Progress in and promise of bacterial quorum sensing research. Nature 551(7680):313–320. https://doi.org/10.1038/nature24624
35. Ali L, Goraya MU, Arafat Y, Ajmal M, Chen JL, Yu D (2017) Molecular mechanism of quorum-sensing in *Enterococcus faecalis*: its role in virulence and therapeutic approaches. Inter J Mol Sci 18(5):e960. https://doi.org/10.3390/ijms18050960
36. Waters CM, Bassler BL (2005) Quorum sensing: cell-to-cell communication in bacteria. Annu Rev Cell Dev Bio 21:319–346. https://doi.org/10.1146/annurev.cellbio.21.012704.131001
37. Teixeira N, Varahan S, Gorman MJ, Palmer KL, Zaidman-Remy A, Yokohata R, de Fátima Silva Lopes, M, (2013) Drosophila host model reveals new *Enterococcus faecalis* quorum-sensing associated virulence factors. PLoS ONE 8(5):e64740
38. De Kievit TR, Iglewski BH (2000) Bacterial quorum sensing in pathogenic relationships. Infect Immun 68(9):4839–4849. https://doi.org/10.1128/IAI.68.9.4839-4849.2000
39. Ali SA, Bin-Asif H, Hasan KA, Rehman M, Abbasi A (2017) Molecular assessment of virulence determinants, hospital associated marker (IS16gene) and prevalence of antibiotic resistance in soil borne Enterococcus species. 2017. Microb Pathog 105:298–306. https://doi.org/10.1016/j.micpath.2017.02.041
40. Zahid S, Bin-Asif H, Hasan KA, Rehman M, Ali SA (2017) Prevalence and genetic profiling of tetracycline resistance (Tet-R) genes and transposable element (Tn916) in environmental *Enterococcus* species. Microb Pathog 111:252–261. https://doi.org/10.1016/j.micpath.2017.09.009
41. Hasan KA, Ali SA, Rehman M, Bin-Asif H, Zahid S (2018) The unravelled Enterococcus faecalis zoonotic superbugs: Emerging multiple resistant and virulent lineages isolated from poultry environment. Zoonoses Public Health 65(8):921–935. https://doi.org/10.1111/zph.12512
42. Rehman M, Hasan KA, Bin-Asif H, Akram S, Zahid S, Mirani ZA, Ali SA (2021) Differentiating *Enterococcus* lineages in combined sewer overflow and potable water combating to hospital acquired high-level β-lactam resistance. Environ Chall 4:e100094. https://doi.org/10.1016/j.envc.2021.100094
43. Andreou LV (2013) Preparation of genomic DNA from bacteria. Methods in Enzymol 529:143–151. https://doi.org/10.1016/B978-0-12-418687-3.00011-2
44. Bin-Asif H, Ali SA (2019) The genus Enterococcus and its associated virulent factors. In: Microorganisms. Blumenberg M, Shaaban M, and Elgaml A Eds. Chapter 6. pp 255–280. IntechOpen Publisher, London, U.K
45. Liu ZL, Chen YJ, Meng QL, Zhang X, Wang XL (2023) Progress in the application of *Enterococcus faecium* in animal husbandry. Front Cell Infect Microbiol 13:e1168189. https://doi.org/10.3389/fcimb.2023.1168189

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